Mastery in Data Analytics and Visualizations and Career Advancement

Lab sheet 03

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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import statsmodels.api as sm

# 1. LOAD DATASET
df = pd.read_csv("WA_Fn-UseC_-HR-Employee-Attrition.csv")

print("=" * 70)
print("STATISTICAL ANALYSIS AND HYPOTHESIS TESTING")
print("=" * 70)

# 2. DISPLAY DATA
print("\nFirst 5 Records:")
print(df.head())

print("\nDataset Dimensions:")
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])
```

```
# 3. DESCRIPTIVE STATISTICS
print("\nDescriptive Statistics:")

numeric_columns = [
    "MonthlyIncome",
    "YearsAtCompany",
    "JobSatisfaction",
    "Age"
]

for column in numeric_columns:
    if column in df.columns:
        print("\n", column)
        print("Mean:", df[column].mean())
        print("Median:", df[column].median())
        print("Mode:", df[column].mode()[0])
        print("Variance:", df[column].var())
        print("Standard Deviation:", df[column].std())

# 4. CORRELATION ANALYSIS
print("\nPearson Correlation:")

correlation_columns = [
    "Age",
    "MonthlyIncome",
    "YearsAtCompany",
    "JobSatisfaction"
```

```
]

correlation = df[correlation_columns].corr(method="pearson")

print(correlation)

plt.figure(figsize=(8, 6))

sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Matrix")
plt.show()

# 5. HYPOTHESIS TESTING
print("\nIndependent Sample T-Test")

# H0: No significant difference in monthly income
# H1: Significant difference in monthly income

stay = df[df["Attrition"] == "No"]["MonthlyIncome"]
leave = df[df["Attrition"] == "Yes"]["MonthlyIncome"]

t_stat, p_value = stats.ttest_ind(
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    stay,
    leave,
    equal_var=False
)

print("T-statistic:", t_stat)
print("P-value:", p_value)

if p_value < 0.05:
    print("Reject H0: Significant difference exists.")
else:
    print("Fail to reject H0: No significant difference found.")

# 6. ONE-WAY ANOVA
print("\nOne-Way ANOVA")

groups = [
    group["JobSatisfaction"].dropna()
    for name, group in df.groupby("JobLevel")
]

f_stat, anova_p = stats.f_oneway(*groups)

print("F-statistic:", f_stat)
print("P-value:", anova_p)

if anova_p < 0.05:
    print("Reject H0: Significant difference exists among groups.")
```

```
else:
    print("Fail to reject H0: No significant difference among groups.")

# 7. LINEAR REGRESSION
print("\nLinear Regression")

X = df["YearsAtCompany"]
y = df["MonthlyIncome"]

X = sm.add_constant(X)

model = sm.OLS(y, X).fit()
print(model.summary())

print("\nRegression Coefficients:")
print(model.params)
print("\nR-squared:", model.rsquared)
# 8. REGRESSION PLOT
plt.figure(figsize=(8, 5))
sns.regplot(
    data=df,
    x="YearsAtCompany",
    y="MonthlyIncome"
)
plt.title("Years at Company vs Monthly Income")
plt.xlabel("Years at Company")
plt.ylabel("Monthly Income")
```

```
plt.show()


# 9. FINAL OBSERVATIONS
print("\n" + "=" * 70)
print("KEY OBSERVATIONS")
print("=" * 70)

print("\n1. T-Test P-value:", p_value)

if p_value < 0.05:
    print("Monthly income differs significantly between employees who leave and stay.")
else:
    print("No statistically significant difference in monthly income was found.")

print("\n2. ANOVA P-value:", anova_p)
if anova_p < 0.05:
    print("Job satisfaction differs significantly among job-level groups.")
else:
    print("No statistically significant difference was found among job-level groups.")
print("\n3. Regression R²:", model.rsquared)
print("\nStatistical analysis completed successfully!")

# Questions and Answers
# 1. What is the difference between descriptive statistics and inferential statistics?
# Answer:
# Descriptive statistics summarize and describe the main features of collected data using measures such as mean, median, mode, variance, and standard deviation. Inferential statistics use sample data to make conclusions or predictions about a larger population.
```

2. Explain the concepts of the Null Hypothesis (H_o) and Alternative Hypothesis (H_1).

Answer:

The Null Hypothesis (H_o) states that there is no significant difference or relationship between variables. The Alternative Hypothesis (H_1) states that a significant difference or relationship exists. Statistical tests are used to decide whether H_o should be rejected.

3. What is a p-value? How is it used to make statistical decisions?

Answer:

A p-value indicates how likely the observed result is if the null hypothesis is true. Generally, if $p < 0.05$, H_o is rejected and the result is considered statistically significant. If $p \geq 0.05$, H_o is not rejected.

4. Differentiate between a t-test and ANOVA. In which situations is each test applied?

Answer:

A t-test is generally used to compare the means of two groups. ANOVA is used to compare the means of three or more groups. For example, a t-test can compare income between employees who leave and stay, while ANOVA can compare satisfaction across multiple groups.

5. What does the Pearson correlation coefficient indicate? What are its possible values?

Answer:

Pearson correlation measures the strength and direction of the linear relationship between two numerical variables. Its value ranges from -1 to +1. A positive value indicates a positive relationship, while a negative value indicates a negative relationship. A value near zero indicates little or no linear relationship.

6. Explain the significance of R^2 (Coefficient of Determination) in Linear Regression.

Answer:

R^2 represents the proportion of variation in the dependent variable that is explained by the independent variable(s). Its value generally ranges from 0 to 1. A higher R^2 indicates that the regression model explains more of the variation in the dependent variable.

7. Why is statistical analysis important before applying machine learning algorithms?

Answer:

Statistical analysis helps understand the data, identify relationships, detect anomalies, check assumptions, and select useful variables. This improves data quality and helps build more reliable machine learning models.

8. What assumptions should be satisfied before performing a t-test or ANOVA?

Answer:

Important assumptions include independent observations, approximately normally distributed data within groups, and similar variances between groups. For ANOVA, the observations should also be independent and the groups should be appropriately defined.

9. How can regression analysis help organizations in forecasting and decision-making?

Answer:

Regression analysis identifies relationships between variables and can be used to predict future outcomes. Organizations can use these predictions for planning, forecasting revenue or costs, estimating demand, and making data-driven decisions.

10. Give two real-world applications where hypothesis testing is commonly used in data analytics.

Answer:

Employee analytics: Testing whether employee income differs significantly between employees who leave and those who stay.

Business analytics: Testing whether different groups, products, or strategies produce significantly different results.

OUTPUT

PS D:\PROGRAMS\Career advancement lab\lab sheet 3> python -u "d:\PROGRAMS\Career advancement lab\lab sheet 3\exp3.py"

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STATISTICAL ANALYSIS AND HYPOTHESIS TESTING

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First 5 Records:

	Age	Attrition	YearsAtCompany	JobSatisfaction	JobLevel	MonthlyIncome
0	60	Yes	13	2	3	11194
1	50	No	6	1	4	11551
2	36	No	8	2	4	11597
3	29	Yes	14	4	1	7926
4	42	No	14	4	1	8475

Dataset Dimensions:

Rows: 50

Columns: 6

Descriptive Statistics:

MonthlyIncome

Mean: 10175.54

Median: 10426.0

Mode: 4229

Variance: 7598370.539183673

Standard Deviation: 2756.5142007948507

YearsAtCompany

Mean: 9.86

Median: 9.5

Mode: 6

Variance: 32.082040816326526

Standard Deviation: 5.664101059861708

JobSatisfaction

Mean: 2.46

Median: 2.0

Mode: 2

Variance: 1.2330612244897956

Standard Deviation: 1.1104328995890727

Age

Mean: 40.96

Median: 42.0

Mode: 23

Variance: 125.83510204081632

Standard Deviation: 11.217624616683175

Pearson Correlation:

	Age	MonthlyIncome	YearsAtCompany	JobSatisfaction
Age	1.000000	0.221271	0.081815	-0.159052
MonthlyIncome	0.221271	1.000000	0.588491	-0.212390
YearsAtCompany	0.081815	0.588491	1.000000	-0.038223
JobSatisfaction	-0.159052	-0.212390	-0.038223	1.000000

Independent Sample T-Test

T-statistic: -1.5623467827624877

P-value: 0.13057375685560213

Fail to reject H0: No significant difference found.

One-Way ANOVA

F-statistic: 0.8250329787750742

P-value: 0.48684647111926926

Fail to reject H0: No significant difference among groups.

Linear Regression

OLS Regression Results

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Dep. Variable: MonthlyIncome R-squared: 0.346

Model: OLS Adj. R-squared: 0.333

Method: Least Squares F-statistic: 25.43

Date: Fri, 28 Aug 2026 Prob (F-statistic): 6.97e-06

Time: 14:44:29 Log-Likelihood: -455.90

No. Observations: 50 AIC: 915.8

Df Residuals: 48 BIC: 919.6

Df Model: 1

Covariance Type: nonrobust

	coef	std err	t	P>|t|	[0.025	0.975]
const	7351.6595	644.187	11.412	0.000	6056.434	8646.885
YearsAtCompany	286.3976	56.792	5.043	0.000	172.209	400.586

Omnibus: 5.477 Durbin-Watson: 2.078

Prob(Omnibus): 0.065 Jarque-Bera (JB): 3.426

Skew: -0.458 Prob(JB): 0.180

Kurtosis: 2.103 Cond. No. 23.1

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Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Regression Coefficients:

const 7351.659508

YearsAtCompany 286.397616

dtype: float64

R-squared: 0.34632218683395866

KEY OBSERVATIONS

1. T-Test P-value: 0.13057375685560213

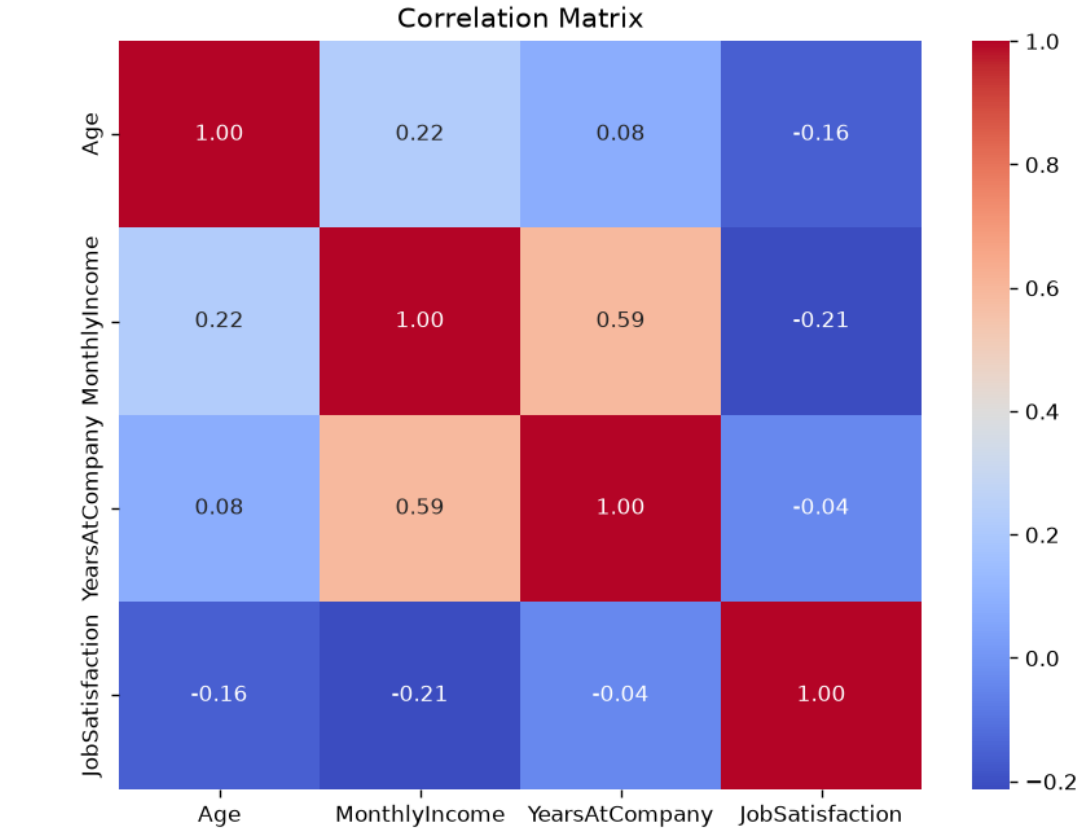
No statistically significant difference in monthly income was found.

2. ANOVA P-value: 0.48684647111926926

No statistically significant difference was found among job-level groups.

3. Regression R^2: 0.34632218683395866

Statistical analysis completed successfully!


Correlation Matrix
Age
MonthlyIncome
YearsAtCompany
JobSatisfaction
1.00
0.22
0.08
-0.16
0.22
1.00
0.59
-0.21
0.08
0.59
1.00
-0.04
-0.16
-0.21
-0.04
1.00
Age
MonthlyIncome
YearsAtCompany
JobSatisfaction
1.0
0.8
0.6
0.4
0.2
0.0
−0.2


Years at Company vs Monthly Income
Monthly Income
16000
14000
12000
10000
8000
6000
4000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Years at Company